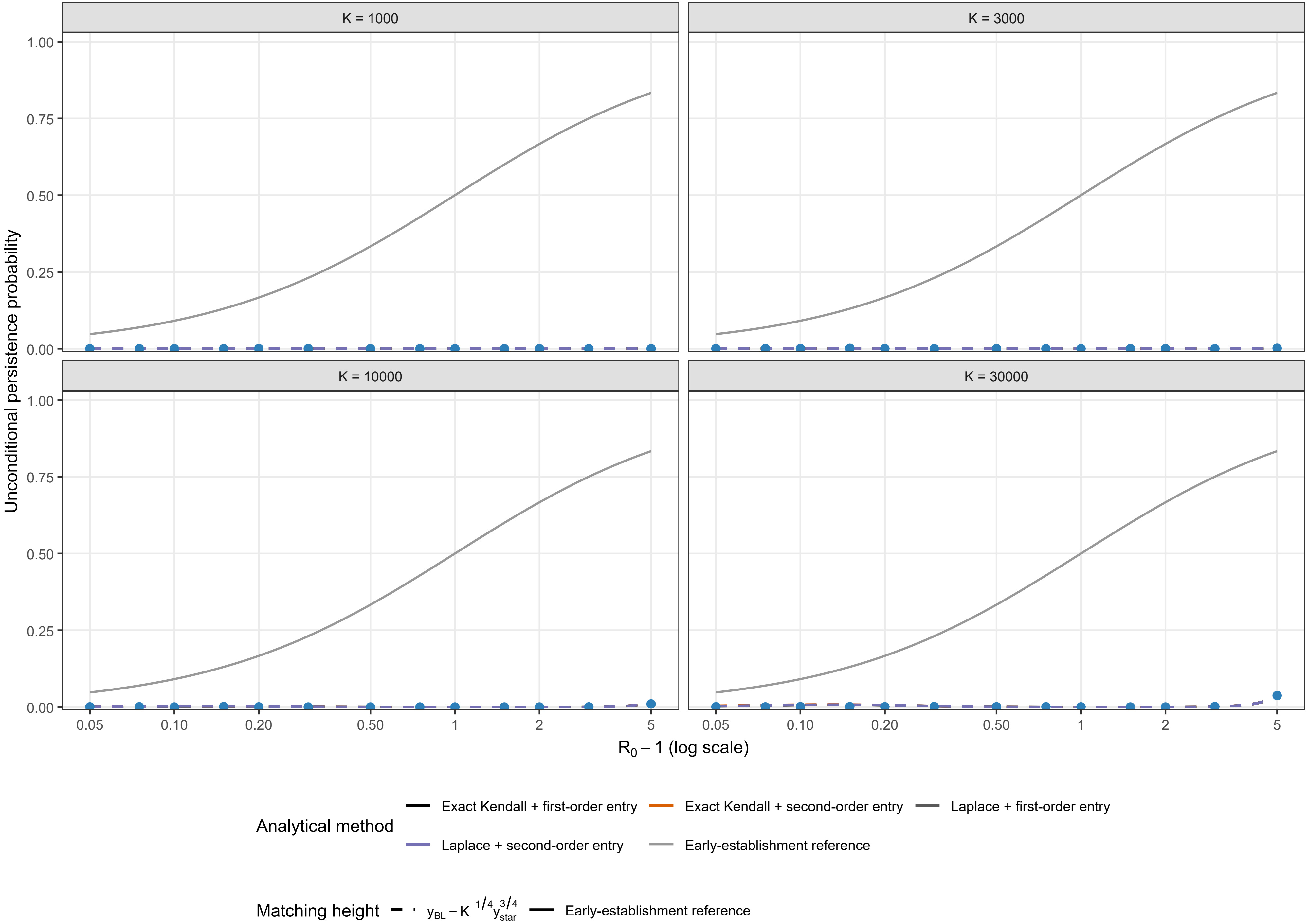
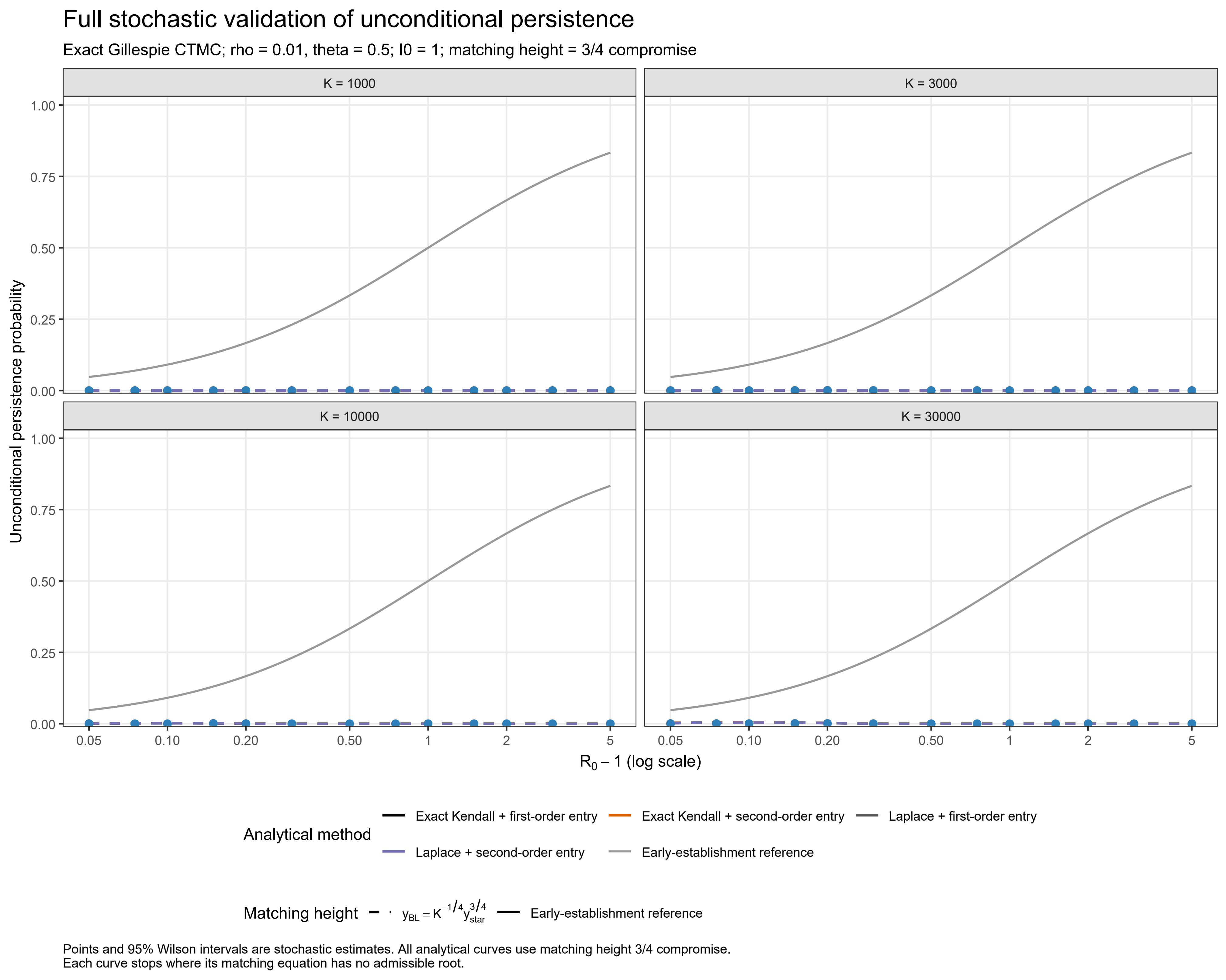
Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; rho = 0.01, theta = 0; I0 = 1; matching height = 3/4 compromise



Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height 3/4 compromise.
Each curve stops where its matching equation has no admissible root.


Full stochastic validation of unconditional persistence
Exact Gillespie CTMC; rho = 0.01, theta = 0.5; I0 = 1; matching height = 3/4 compromise
K = 1000
K = 3000
K = 10000
K = 30000
Unconditional persistence probability
0.00
0.25
0.50
0.75
1.00
0.05
0.10
0.20
0.50
1
2
5
$R_0 - 1$ (log scale)
Analytical method
Exact Kendall + first-order entry
Exact Kendall + second-order entry
Laplace + first-order entry
Laplace + second-order entry
Early-establishment reference
Matching height
$y_{BL} = K^{-1/4} y_{star}^{3/4}$
Early-establishment reference
Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height 3/4 compromise.
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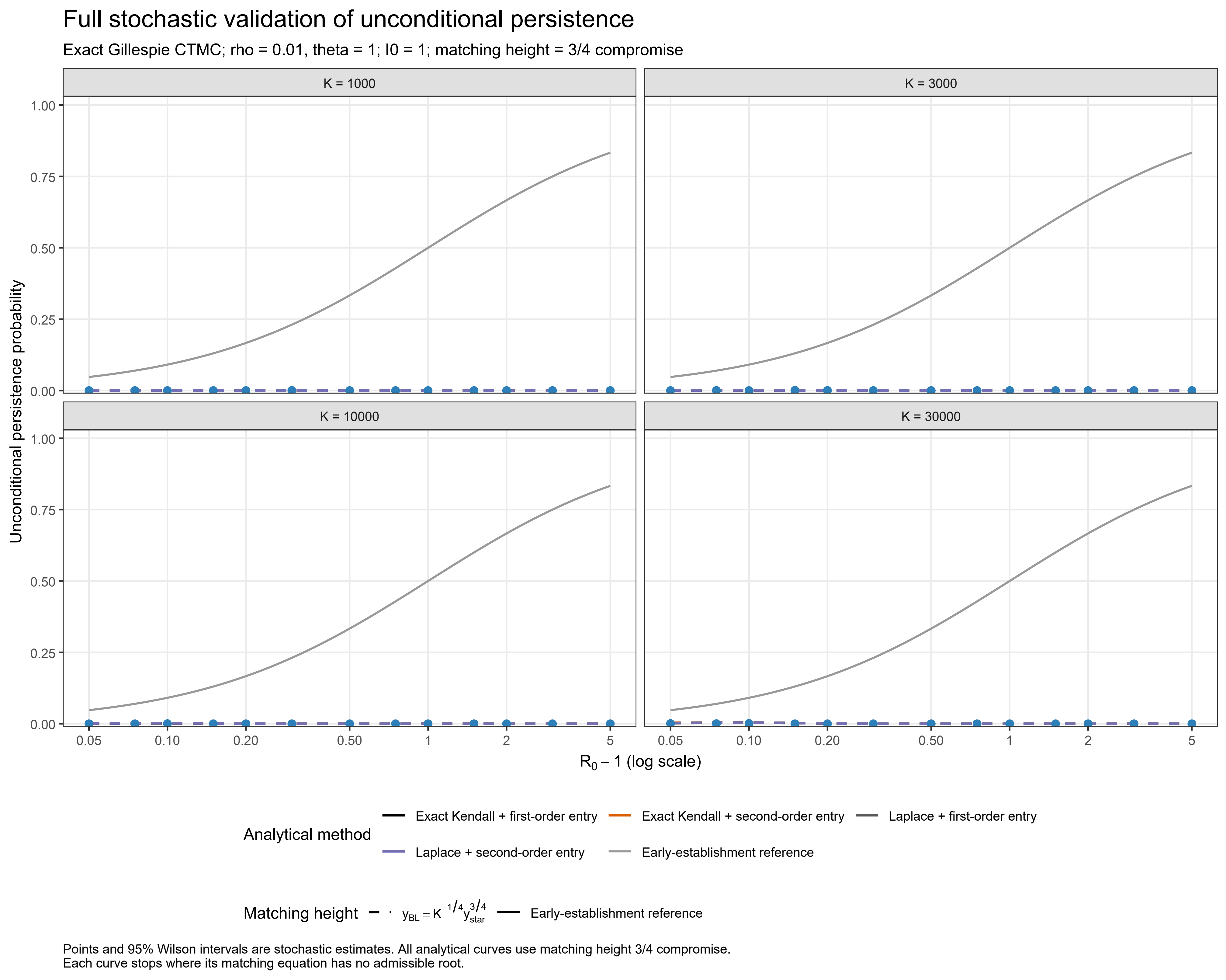

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K = 3000
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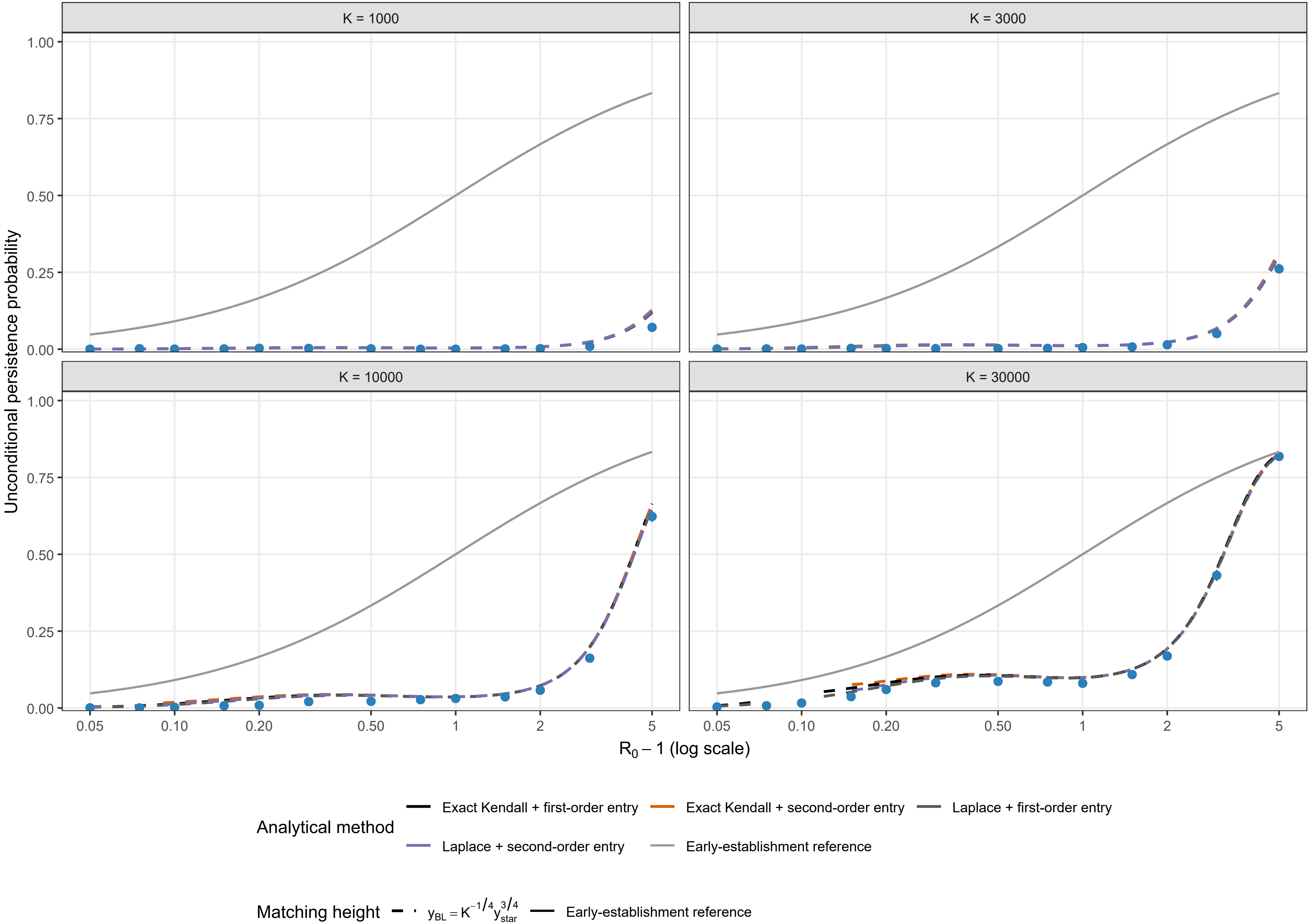
Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; rho = 0.02, theta = 0; I0 = 1; matching height = 3/4 compromise



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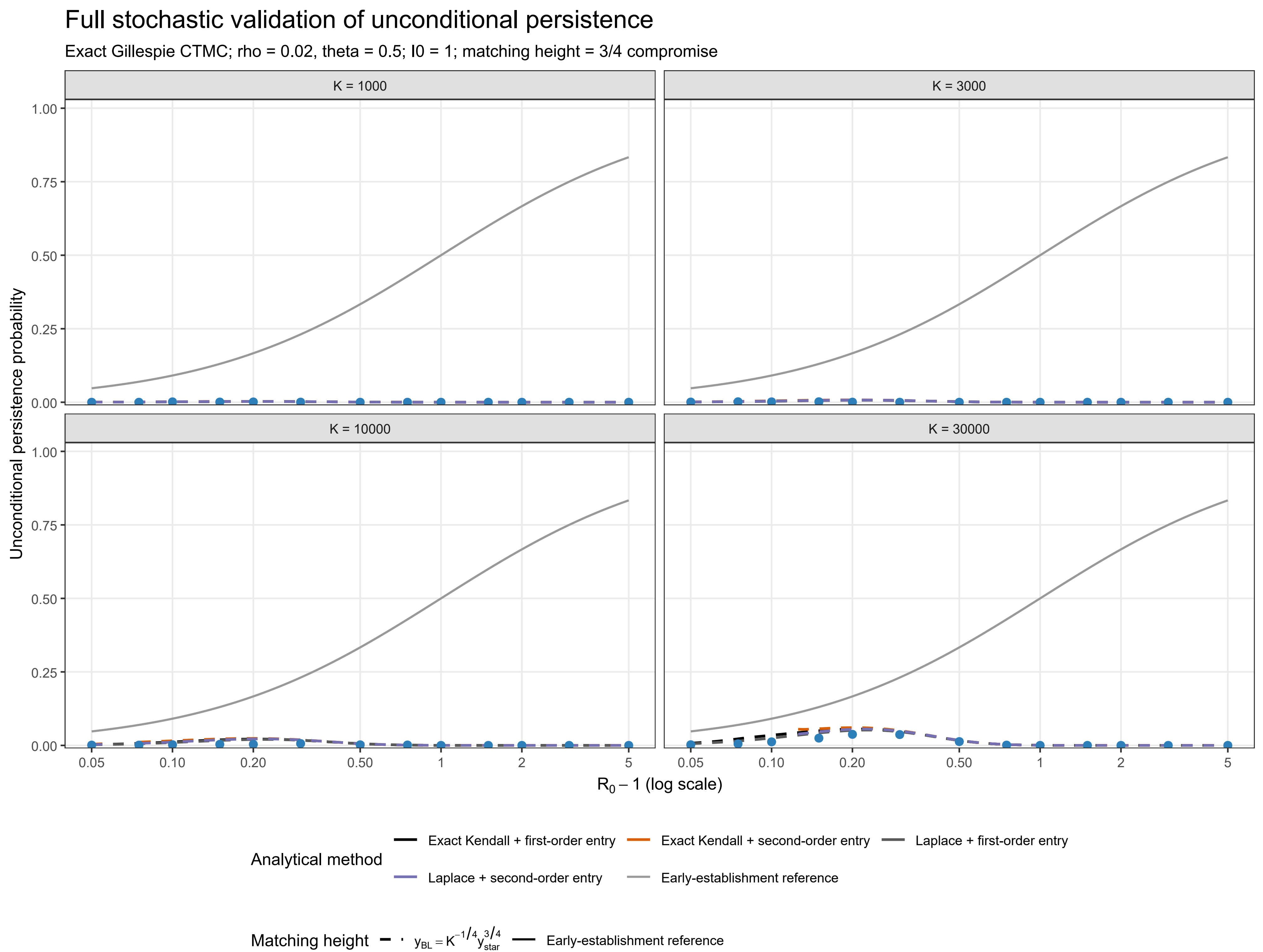

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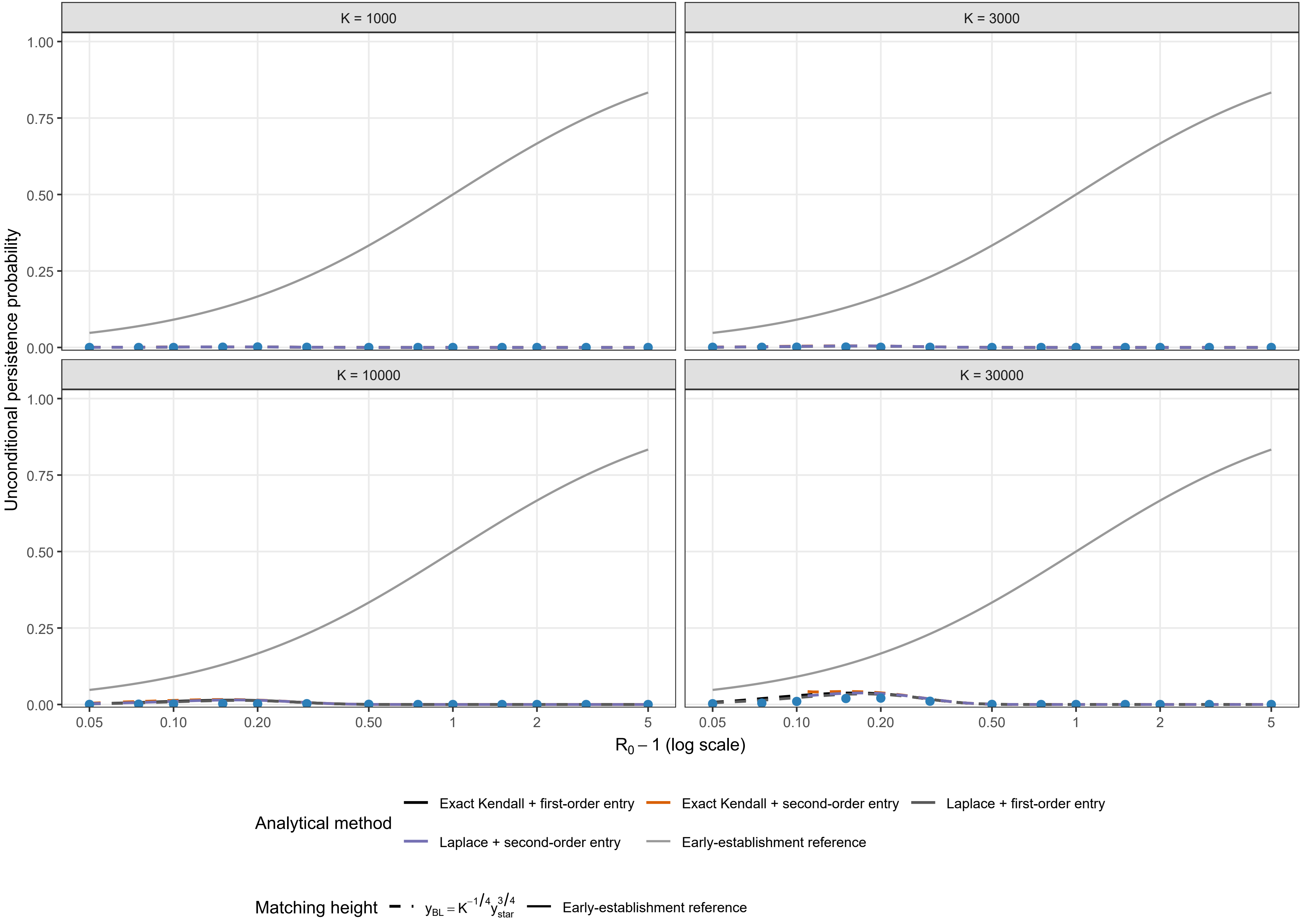
Full stochastic validation of unconditional persistence

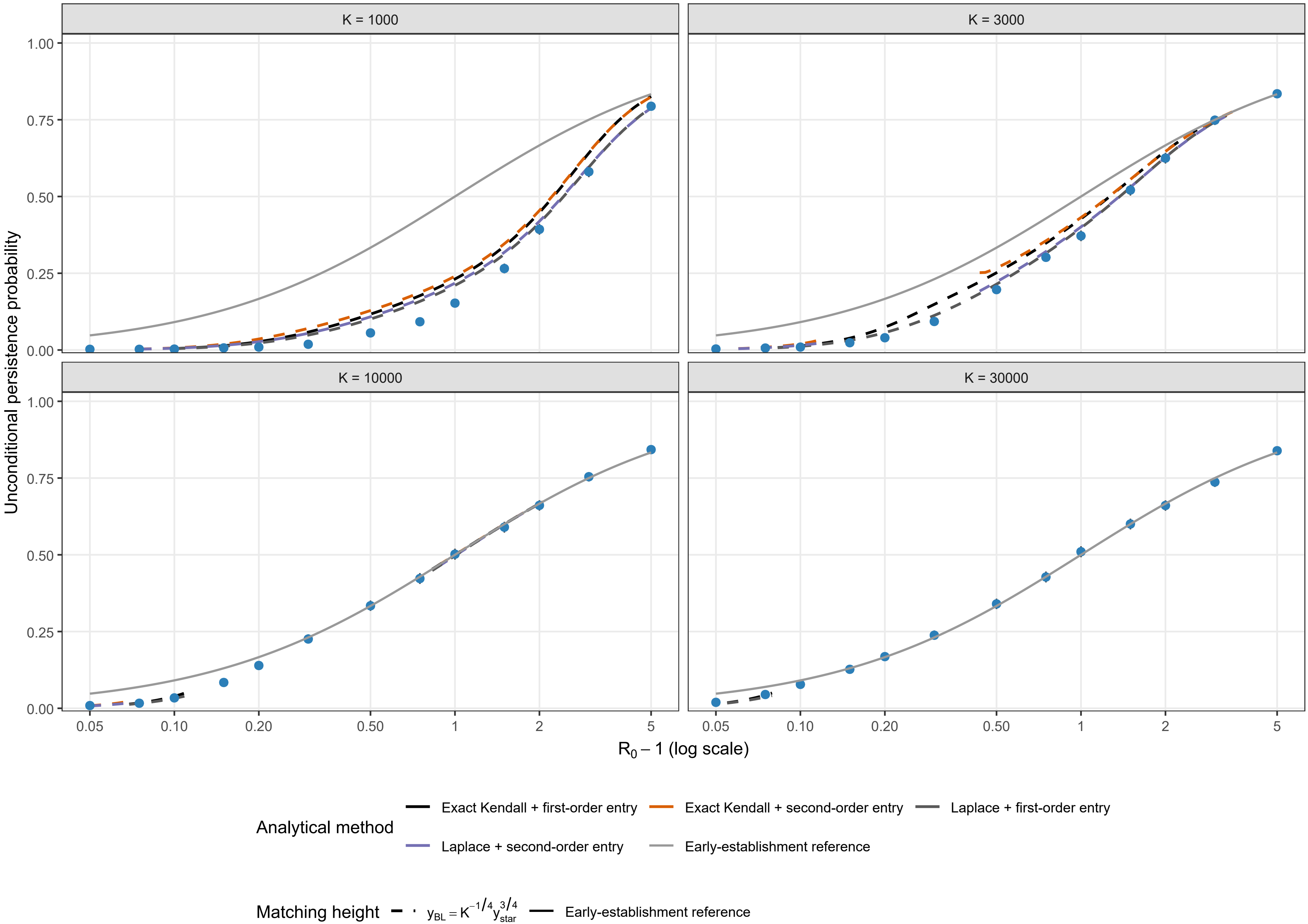
Exact Gillespie CTMC; rho = 0.02, theta = 1; I0 = 1; matching height = 3/4 compromise



Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height 3/4 compromise.
Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; rho = 0.05, theta = 0; I0 = 1; matching height = 3/4 compromise



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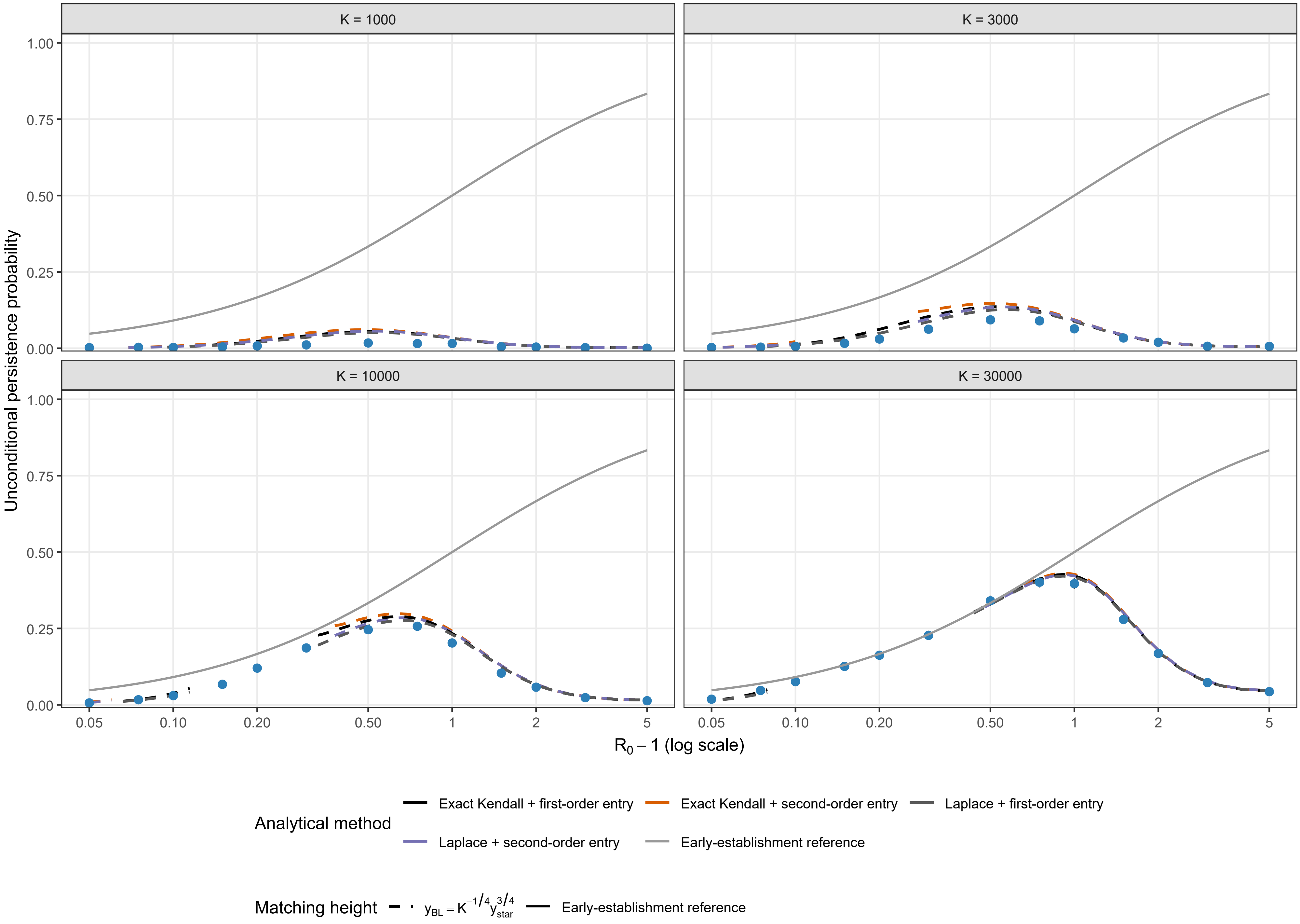

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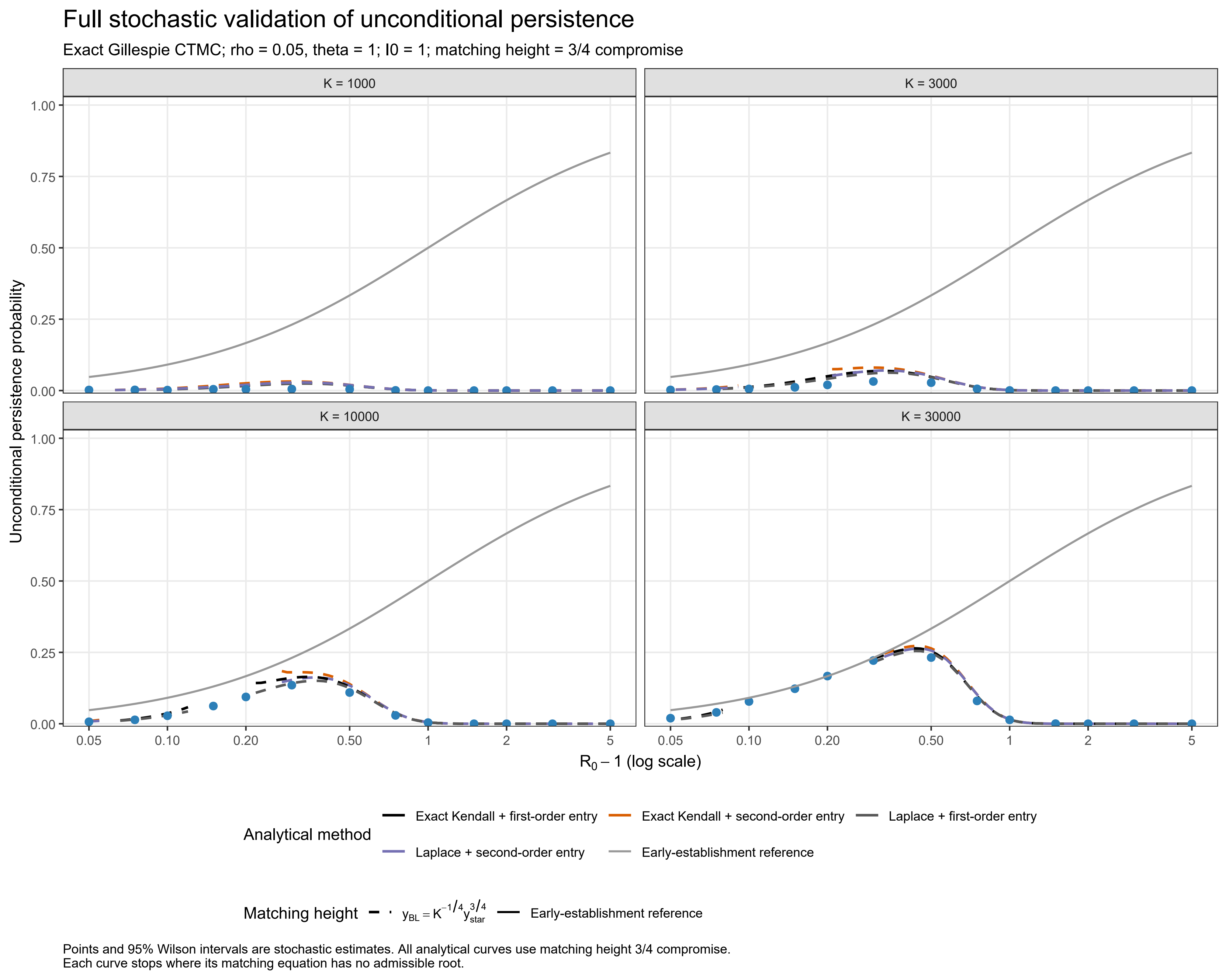

Full stochastic validation of unconditional persistence
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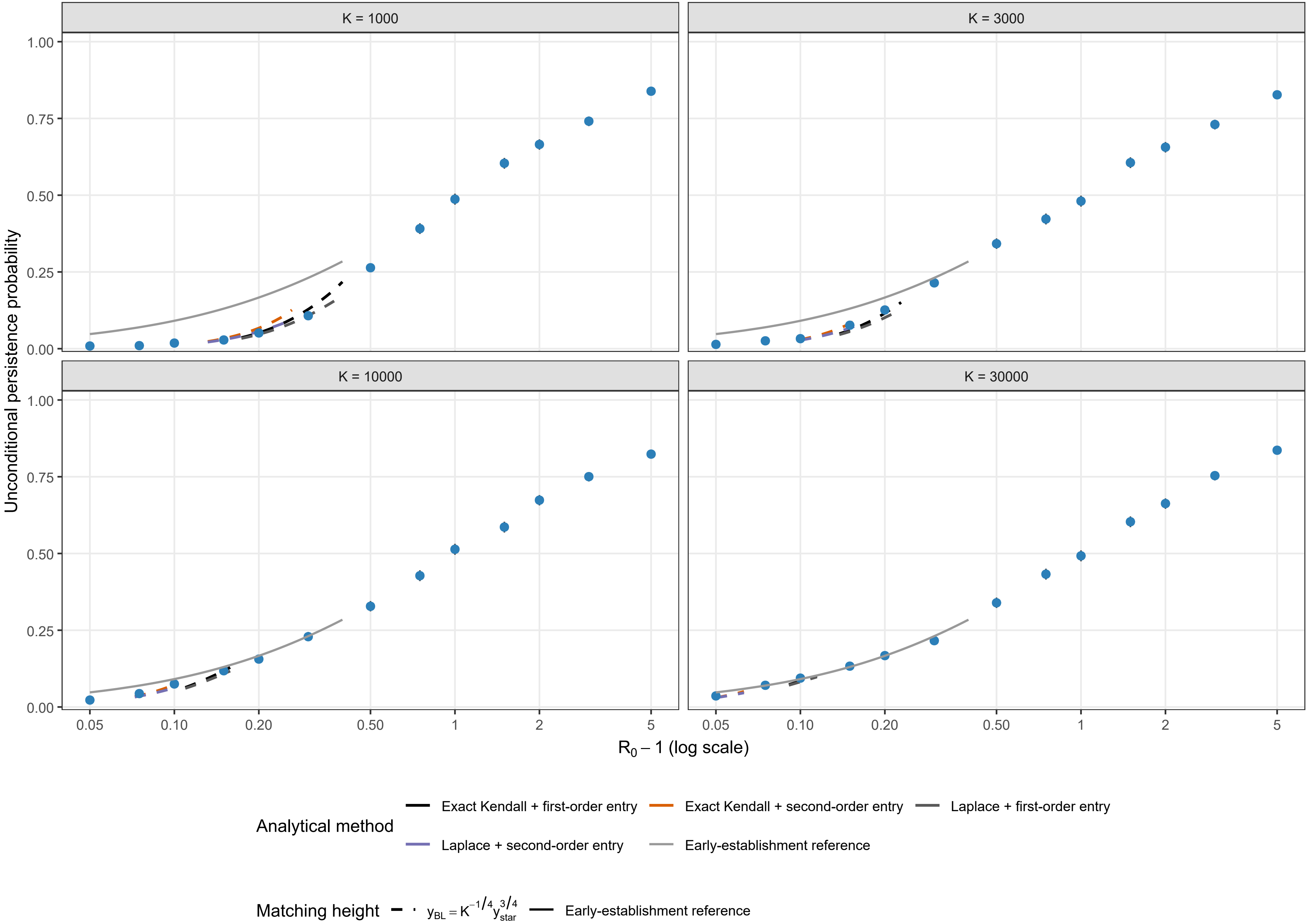
Early-establishment reference
Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height 3/4 compromise.
Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

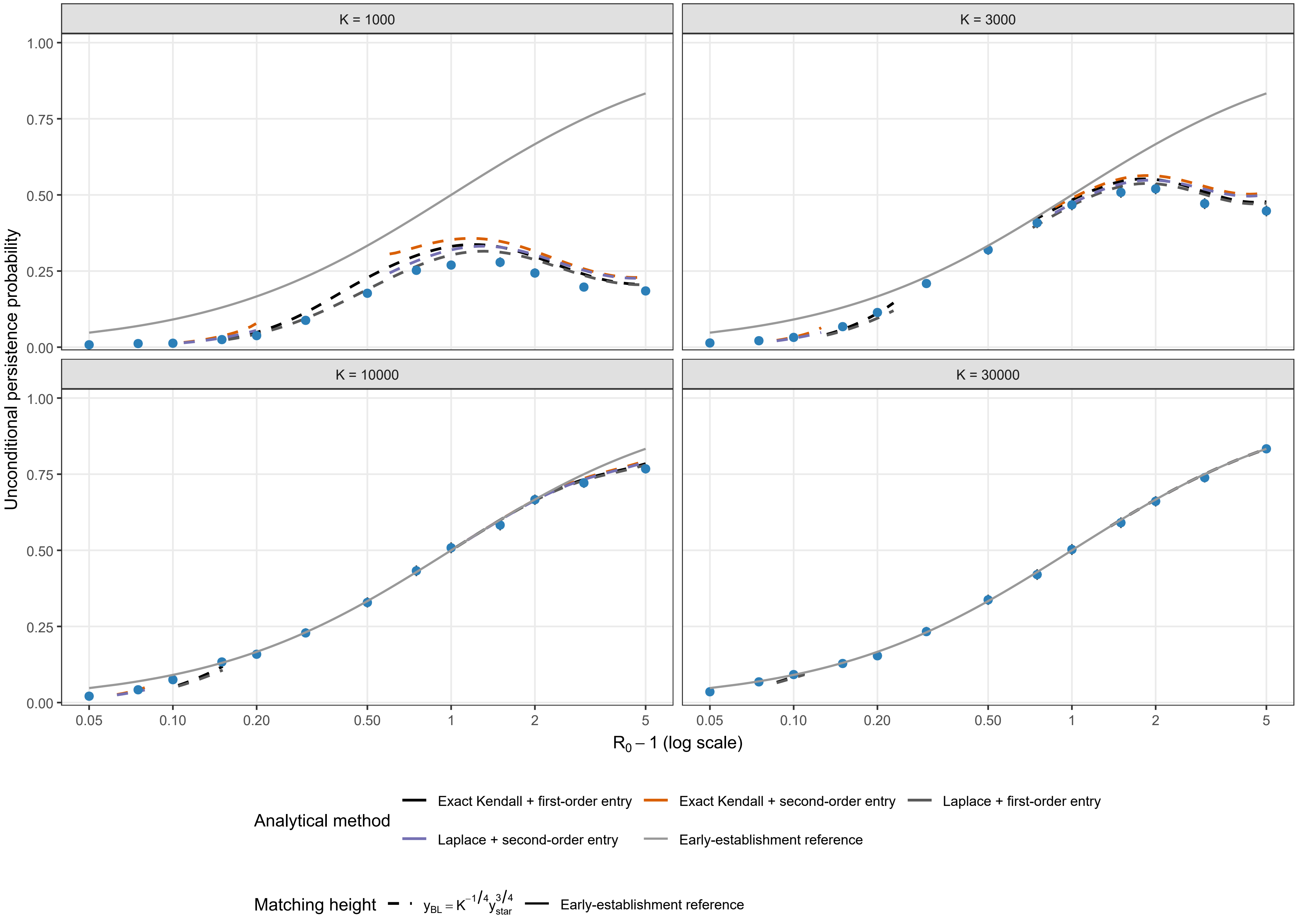
Exact Gillespie CTMC; rho = 0.10, theta = 0; I0 = 1; matching height = 3/4 compromise



Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height 3/4 compromise.
Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; rho = 0.10, theta = 0.5; I0 = 1; matching height = 3/4 compromise



Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height 3/4 compromise.
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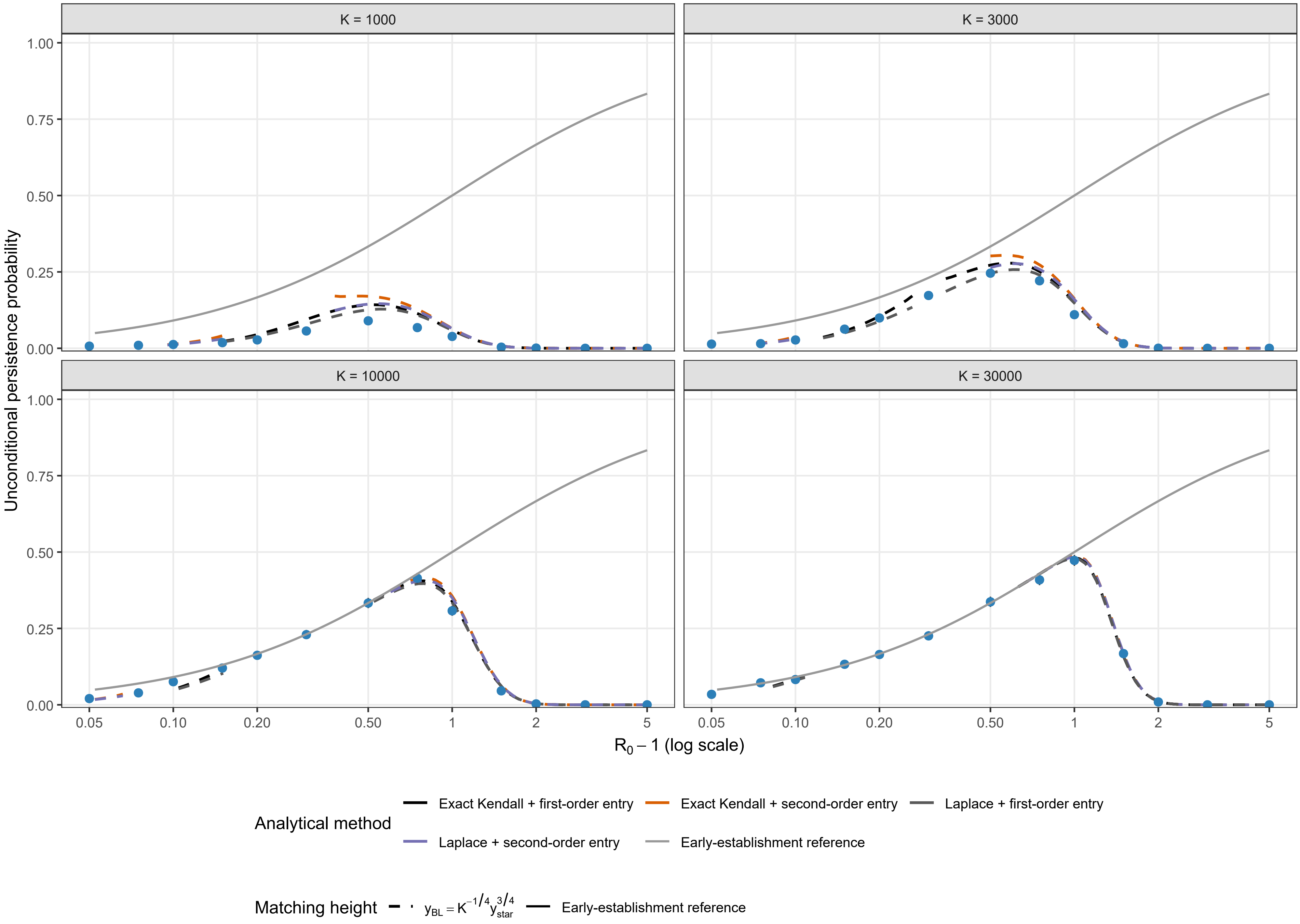
Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; rho = 0.10, theta = 1; I0 = 1; matching height = 3/4 compromise



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